HDOC Day-50

Question

Write a C program to read a variable-length integer array nums[]. Pass it by address to majority_element(). Count occurrences and print an element that appears strictly more than size/2 times; otherwise print -1.

C Program

```
#include <stdio.h>

void majority_element(int *nums, int size) {
    for (int i = 0; i < size; i++) {
        int count = 0;
        for (int j = 0; j < size; j++) {
            if (*(nums + i) == *(nums + j))
                count++;
        }
        if (count > size / 2) {
            printf("%d\n", *(nums + i));
            return;
        }
    }
    printf("-1\n");
}

int main() {
    int size;
    scanf("%d", &size);
    if (size <= 0) return 0;
    int nums[size];
    for (int i = 0; i < size; i++)
        scanf("%d", &nums[i]);
    majority_element(nums, size);
    return 0;
}
```

Sample Input and Output

```
Input:
5
3 3 2 3 3

Output:
3
```

Explanation

The function counts how often each value appears. If its count is greater than half the array size, it is the majority element.